

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

Code of Conduct:

I D.Mani Kumar Redddy(192325095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

D.Mani Kumar Reddy

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1.An image contains high-frequency noise.Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

ANSWER:

The most suitable filter is the Low Pass Filter (LPF) because high-frequency noise is removed while preserving the important low-frequency information.

Working:

- Convert the image into the frequency domain using the Fourier Transform.
- Apply a Low Pass Filter to allow only low-frequency components to pass.
- High-frequency components (noise) are blocked.
- Convert the filtered image back to the spatial domain using the Inverse Fourier Transform.

Impact on Image Quality:

- Removes random noise effectively.
- Produces a smoother image.
- Slightly blurs edges and fine details.
- Improves the overall visual appearance.

Conclusion:

A Low Pass Filter is best for removing high-frequency noise while maintaining the important information in the image.

2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

ANSWER:

The suitable technique is the Ideal Low Pass Filter (ILPF).

Working:

- Transform the image into the frequency domain.
- Set a cutoff frequency.
- Frequencies below the cutoff are kept.
- Frequencies above the cutoff are removed.
- Apply the Inverse Fourier Transform to obtain the filtered image.

Advantages:

- Preserves low-frequency information.
- Removes high-frequency noise.
- Produces a smooth image.

Disadvantages:

- May create ringing artifacts near edges.
- Slight loss of image sharpness.

Conclusion:

The Ideal Low Pass Filter effectively preserves useful image information while reducing unwanted noise.

3.An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

ANSWER:

The appropriate filter is the High Pass Filter (HPF).

Working:

- Convert the image to the frequency domain.
- Remove low-frequency components.
- Keep high-frequency components.
- Convert the image back using the Inverse Fourier Transform.

Effect on Frequency Components:

- Enhances edges and fine details.
- Removes slowly changing intensity values.
- Makes boundaries more visible.

Advantages:

- Improves image sharpness.
- Highlights object edges.
- Useful for feature extraction.

Conclusion:

The High Pass Filter enhances image edges by preserving high-frequency information.

4.A smooth version of an image is required.Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

ANSWER:

The Gaussian Low Pass Filter (GLPF) is commonly used for image smoothing.

Working:

- Convert the image into the frequency domain.
- Apply the Gaussian filter.
- Gradually reduce high-frequency components.
- Convert back using the Inverse Fourier Transform.

Results:

- Smooths the image.
- Removes noise.
- Produces fewer ringing effects than the Ideal LPF.
- Preserves image quality better.

Advantages:

- Smooth transition.
- Better visual quality.
- Effective noise reduction.

Conclusion:

The Gaussian Low Pass Filter produces a smooth image with less distortion and fewer artifacts.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

ANSWER:

The appropriate technique is Threshold Segmentation.

Working:

- Select a threshold value.
- Pixels greater than the threshold become the object.

- Pixels below the threshold become the background.

Advantages:

- Very simple.
- Fast.
- Accurate when the object is brighter than the background.

Applications:

- Document processing.
- Medical imaging.
- Industrial inspection.

Conclusion:

Threshold segmentation effectively separates bright objects from dark backgrounds.

6.You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

ANSWER:

The suitable method is Global Thresholding.

Working:

1. Choose a threshold value.
2. Compare every pixel with the threshold.
3. Pixels above the threshold become foreground.
4. Pixels below the threshold become background.

Advantages:

- Easy to implement.
- Fast processing.
- Suitable for images with uniform lighting.

Conclusion:

Global Thresholding separates foreground and background effectively using pixel intensity values.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

ANSWER:

The Canny Edge Detection method is the most suitable.

Working:

1. Smooth the image using a Gaussian filter.
2. Calculate intensity gradients.
3. Apply Non-Maximum Suppression.
4. Use Double Thresholding.
5. Track edges using hysteresis.

Advantages:

- Accurate edge detection.
- Less sensitive to noise.
- Produces thin and continuous edges.

Applications:

- Object detection.
- Image recognition.
- Medical imaging.

Conclusion:

Canny Edge Detection provides accurate and reliable edge detection with minimal noise.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

ANSWER:

The suitable technique is Region Growing.

Working:

- Select a seed pixel.
- Compare neighboring pixels.
- Add pixels with similar intensity.
- Continue until no similar pixels remain.

Advantages:

- Produces connected regions.
- Accurate segmentation.
- Works well when neighboring pixels have similar intensity.

Applications:

- Medical image analysis.
- Satellite image processing.

- Object extraction.

Conclusion:

Region Growing groups neighboring pixels with similar intensity to form meaningful image regions.

9. Detected edges in an image appear broken.

Explain and justify a method to improve edge continuity, including its role in post-processing.

ANSWER:

The suitable method is Morphological Closing.

Working:

- Perform Dilation to connect nearby edges.
- Perform Erosion to restore the original shape.
- Small gaps between edges are filled.

Advantages:

- Connects broken edges.
- Removes small holes.
- Improves object boundaries.

Role in Post-processing:

- Produces continuous edges.
- Makes object detection more accurate.
- Improves segmentation quality.

Conclusion:

Morphological Closing improves edge continuity and enhances the final segmented image.

10.You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

ANSWER:

The appropriate technique is Boundary Extraction using Morphological Operations.

Working:

1. Perform erosion on the binary image.
2. Subtract the eroded image from the original image.
3. The remaining pixels represent the object boundary.

Formula:

Boundary = Original Image – Eroded Image

Advantages:

- Extracts clear object outlines.
- Simple and efficient.
- Preserves the original object shape.

Applications:

- Shape analysis.
- Object recognition.
- Medical image processing.

Conclusion:

Boundary Extraction effectively identifies object outlines by highlighting only the boundary pixels, making object analysis easier.